ISAC 868 MHz LoRaWAN Antenna-Sensor — System Design & Manufacturing Summary (Rev E.5)

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Hardware Revision: Rev E.5 (Hardware Revision E.5 , fab/ISAC_868_RevE5_PCBWay_Turnkey.zip)
Target Radio / MCU: RAKwireless RAK3172LP-SIP-8-SM-NI (STM32WLE5JC + SX1262 RFO_LP , 3.0 V DIO3 TCXO 0x06 , EU868)
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1. Architecture Overview

ISAC_868_KiCad9 (Rev E.5) combines **868 MHz LoRaWAN Class A wireless telemetry** (+14.0 dBm TX with adaptive SF7/SF9/SF12 wet-radome link recovery) and **near-field surface-permittivity sensing** (+4.0 dBm 3-tone CW reflectometry at 863.1 / 866.5 / 869.5 MHz , < 1.25 ms/tone with Listen-Before-Sense ambient RF blanking) on a single 90.00 × 40.00 × 1.60 mm (+0.03/-0.10 mm) 2-layer FR-4 (1.51 mm core, 1 oz ENIG) PCBA (67 populated top-side SMD parts, 0 THT parts). The board mounts inside a flat-roof (Z = +8.50 mm , -1.44 mm compliant skirt), open-East self-draining 0.80 mm UV-stabilized ASA/PETG enclosure (95.7 × 45.7 × 16.1 mm) sealed by a continuous 360° Parker ORD 5700 EPDM O-ring (1.80 × 0.76 mm groove, 23.3% squeeze, 68.3% gland fill) and bonded across the sensing aperture with 25 µm 3M 467MP acrylic transfer adhesive.

1. **Shared IFA + Interdigital Antenna (AE1) & Elliptic Harmonic Pi-Filter:**
2. Synthesized with a 3D Yee-grid FDTD solver (rf/fdtd_3d_ant.c + rf/ifa_enclosure_mom.py , modeling IPC-4552 ENIG NiP/Au sheet resistance R_S = 23.74 mΩ/square and substrate/radome dielectric loss). In enclosed Dry Air, resonance sits at f0 = 869.00 MHz with raw |S11|(866.5) = -23.69 dB (-20.83 dB cascaded), |S11| ≤ -10.21 dB across 863–870 MHz over all three FR-4 permittivity corners (εr ∈ {4.2, 4.4, 4.6}), total efficiency η_tot = 45.5% (η_rad = 50.0%), and 43.73 MHz/RIU Air-to-Ice sensitivity (869.0 MHz → 834.5 MHz).
3. A 3rd-order elliptic low-pass Pi-filter (C_M1 = 3.6 pF GJM1555C1H3R6WB01D , L_M1 = 8.2 nH LQW15AN8N2G00D || C_MZ = 0.5 pF GJM1555C1HR50WB01D , C_M2 = 4.3 pF GJM1555C1H4R3WB01D) attenuates 2f0 (1.733 GHz) by ≥ 37.2 dB and 3f0 (2.600 GHz) by ≥ 60.1 dB across Dry Air and Water VSWR loads (ETSI EN 300 220-1 ≤ -36 dBm).
4. A continuous bare-ENIG F.Mask opening (X_gbr = [73.40..87.80] mm, Y_gbr = [6.65..20.80] mm , no silkscreen in X_gbr ≥ 50.0 mm) avoids solder-mask ridges beneath the bonded 0.80 mm sensing deck.
5. **Asymmetric Microstrip Return-Loss Bridge & 2-Point Calibrated Reflectometer:**
6. Uses a thin-film Yageo RT0402BRD078R2L (8.2 Ω ±0.1%) inline sense resistor R1 , 0.1% Susumu RG1005P reference arms (R2 = 681 Ω, R3 = 3.57 kΩ, R_TERM = 56 Ω), ±2% Murata LQW15AN inductors (L_Q = 27 nH, L_S = 30 nH), ±0.05 pF GJM1555C1H capacitors (C_S = C_S2 = 1.0 pF), and an Analog Devices LT5538IDD#TRPBF logarithmic detector (IN+ complex load 291.65 Ω || 1.194 pF at 866.5 MHz).
7. Placing a 2.20 × 2.20 mm F.Cu + B.Cu ground cutout beneath /BR_S (C_par = 0.1305 pF < 0.15 pF) yields -0.802 dB through loss, 24.81 dB 50 Ω directivity at 866.5 MHz , and 17.60 dB P01 Air-to-Ice contrast across 2,000 Monte-Carlo trials.
8. Running CW sensing at +4.0 dBm keeps LT5538IDD#TRPBF (U2) input ≤ -25.40 dBm (+12.40 dB below -13 dBm compression), while unpowered +14 dBm LoRaWAN uplinks produce ≤ 39.5 mV_pk at U2 IN+ across VSWR 6:1 . Two-point factory calibration (Dry-Air Vout tare + Al2O3 high-εr level reference cal_ref_cdb , paired with 0.015 dB/°C residual thermal compensation, TX-off Listen-Before-Sense jammer blanking, and -9 dBm soft-ramp CW quenching) achieves 99.72% Water vs IPA yield (P01 separation 0.65 dB ≥ 0.50 dB).
9. **Split-Rail Power Tree, Direct-Soldered Battery Pads & Freestanding C11 Firmware:**
10. J2 (B2B-PH-K-S) and debug headers J1, J3, J4, J5 are marked DNP = Yes so turnkey assembly is single-pass top-side SMT (67 SMD parts). During box-build, 24–26 AWG silicone battery leads solder upward from B.Cu into J2 's bare-ENIG Ø0.75 mm PTH pads (Pad 1 Square = GND , Pad 2 Round = +VBAT_RAW , 4.35 V DC MAX , DOWSIL 3145 RTV strain relief).
11. Q3 (AO3401A P-MOSFET ideal diode, < 6 mV drop) feeds U3 (TPS7A0233DBVR 25 nA LDO, C_OUT_eff ≈ 13.8 µF ≤ 22 µF), Q1 (TPS22919DCKR + C_Q1 = 100 nF) powers LT5538 directly from VBAT (keeping U3 peak load at 24.1 mA , 8.30× below 200 mA), L_SMPS = LQH2MCN150K02L (Isat = 300 mA) supports the internal SX1262 buck converter, and Q2 (NX3008CBKS,115) gates battery sensing on PA9 (150 µs window).

12. Freestanding C11 firmware (15,948 B Flash, 408 B RAM) configures DIO3 TCXO to 3.0 V (0x06), holds unused pins (PA2/PA3) in Analog High-Z during STOP2 , enforces LoRaWAN 1.0.3 §7 Join backoff (17,281 s Tier-3) and §5.4 RXParamSetupReq , persists session state in commit-last dual-page Flash NVM with 72-bit ECCD NMI self-healing, and records per-unit UID64 + 2-point calibration in provisioning_vault/ .

2. Summary Specifications (firmware/test/data/design_values.json)

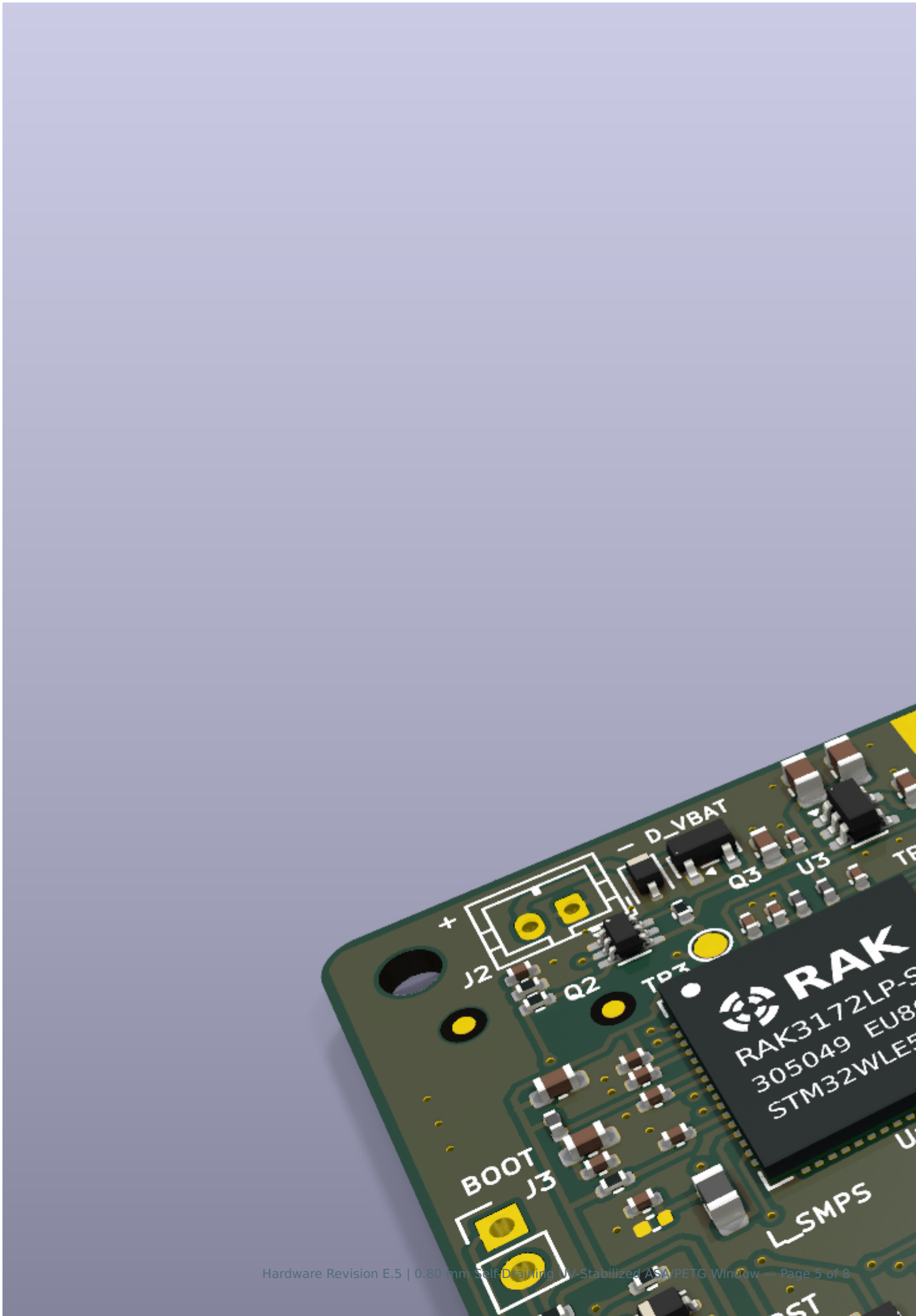
Subsystem Metric	Design / Simulated Value	Target
PCB & Assembly Counts	92 footprints (67 SMD pop, 0 THT; 10 DNP incl. J2 ; 15 BOARD_ONLY), 131 tented vias	0 ERC, 0 DRC, 0 DNP paste
Surface Finish, Mask & CPWG Z0	IPC-4552 ENIG ; 0.035 mm global / 0.025 mm U1,U2 NSMD (≥ 0.180 mm dams); Z0 = 51.90 Ω (51.2–57.1 Ω , RL ≥ 26.8 dB)	ImpedanceControlled = False
Bridge Through Loss & Directivity	|S21| = -0.802 dB , Directivity(866.5) = 24.81 dB (C_par = 0.1305 pF)	≥ 22.0 dB (< 0.15 pF)
2-Point Calibrated 2,000 -Trial MC	Air-to-Ice P01 = 17.60 dB (100% yield); Water vs IPA P01 = 0.65 dB (99.72% yield)	P01 ≥ 8.0 dB / ≥ 0.50 dB
Enclosed AE1 Match & Efficiency	f0 = 869.0 MHz , |S11|(866.5) = -20.83 dB , εr∈{4.2,4.4,4.6} worst |S11| ≤ -10.21 dB , η_tot = 45.5%	|S11| ≤ -10.0 dB all εr
Harmonic Pi-Filter (2f0 / 3f0)	37.20 dB (1.733 GHz) / 60.08 dB (2.600 GHz) enclosed (37.46 / 60.65 dB water)	≥ 28 dB / ≥ 45 dB
Weatherproof Enclosure (95.7 × 45.7 × 16.1 mm)	Flat +8.50 mm roof + 0.80 mm open-East ASA/PETG deck + 25 µm 3M 467MP + Parker ORD 5700 O-ring (23.3% squeeze, 68.3% fill, M2.5 × 22 mm corner screws)	Watertight 2-manifold (0 3D clashes)

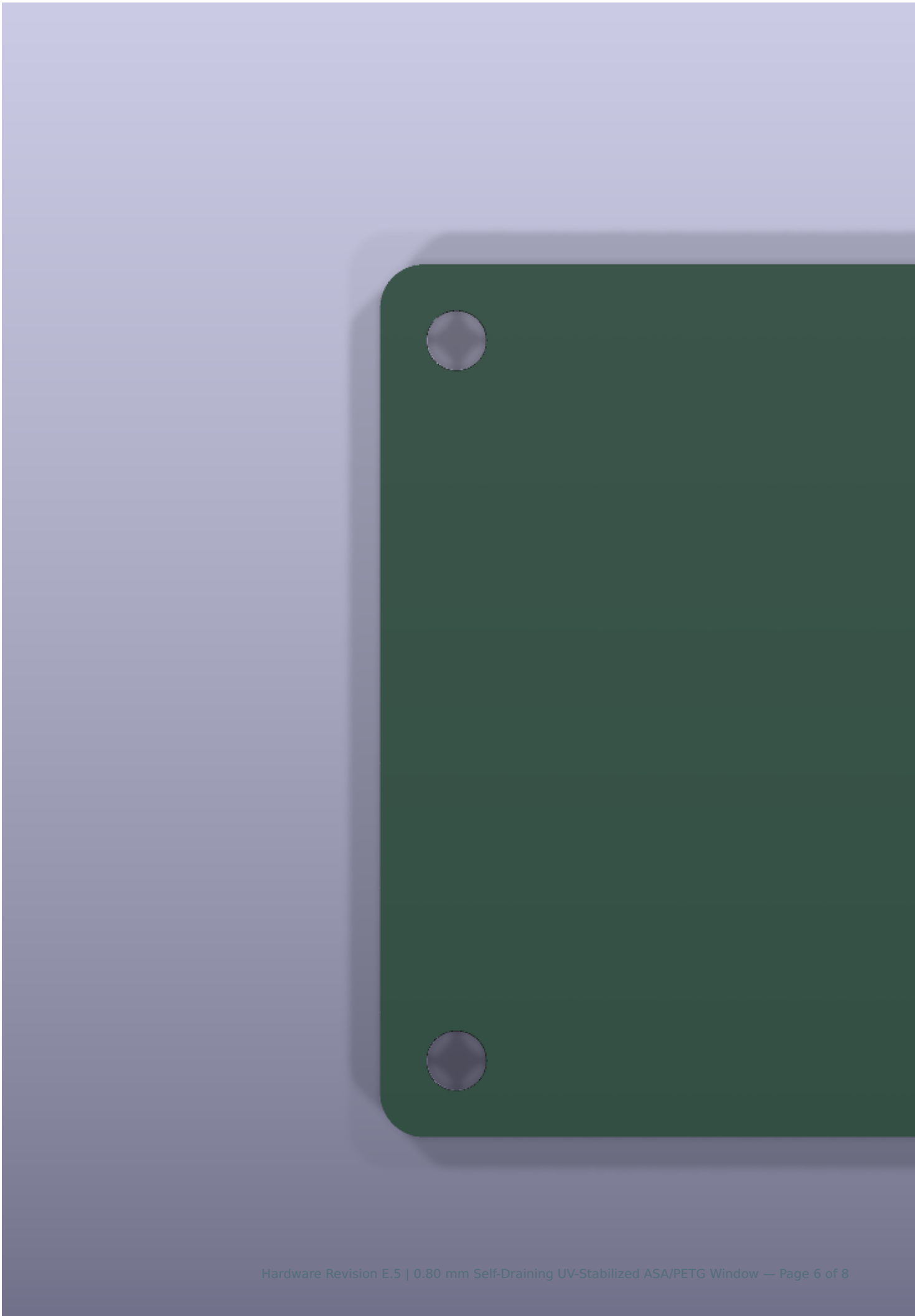
3. 3D FDTD 3-Tone Benchmark Table (+4.0 dBm CW Sensing, 2-Point Calibrated)

External Superstrate (2.0 mm on 0.80 mm ASA/PETG)	ε_r^* (868 MHz)	f_{res} [MHz]	Cascaded | S11| (f1 / f2 / f3) [dB]	P_det (f1 / f2 / f3) [dBm]	Tilt [dB]	Notch [dB]	Level [dBm]	FPort 10 State
Dry Air	1.00 - j0.000	869.0	-18.79 / -20.83 / -18.33	-45.75 / -48.92 / -42.31	-3.44	+4.88	-45.66	0 = DRY_AIR
PTFE / Oil	2.10 - j0.0002	849.0	-8.72 / -7.14 / -5.99	-33.83 / -33.36 / -32.60	-1.23	+0.15	-33.26	1 = OIL
Glaze Ice	3.20 - j0.003	834.5	-3.65 / -2.89 / -2.32	-28.61 / -28.66 / -28.60	-0.01	+0.06	-28.63	2 = GLAZE_ICE
Alumina (Al2O3)	9.80 - j0.002	783.0	-0.36 / -0.53 / -0.69	-25.40 / -26.21 / -26.96	+1.56	+0.03	-26.19	3 = IPA (Cal Ref)
IPA (868 MHz)	6.29 - j6.090	794.0	-2.27 / -2.19 / -2.12	-27.01 / -27.77 / -28.46	+1.45	+0.04	-27.75	3 = IPA
Water Film	79.9 - j3.810	776.0	-0.71 / -0.72 / -0.71	-25.73 / -26.38 / -26.95	+1.22	+0.04	-26.35	4 = LIQUID_WATER

4. CAD & Mechanical Renderings (`Rev E.5`)

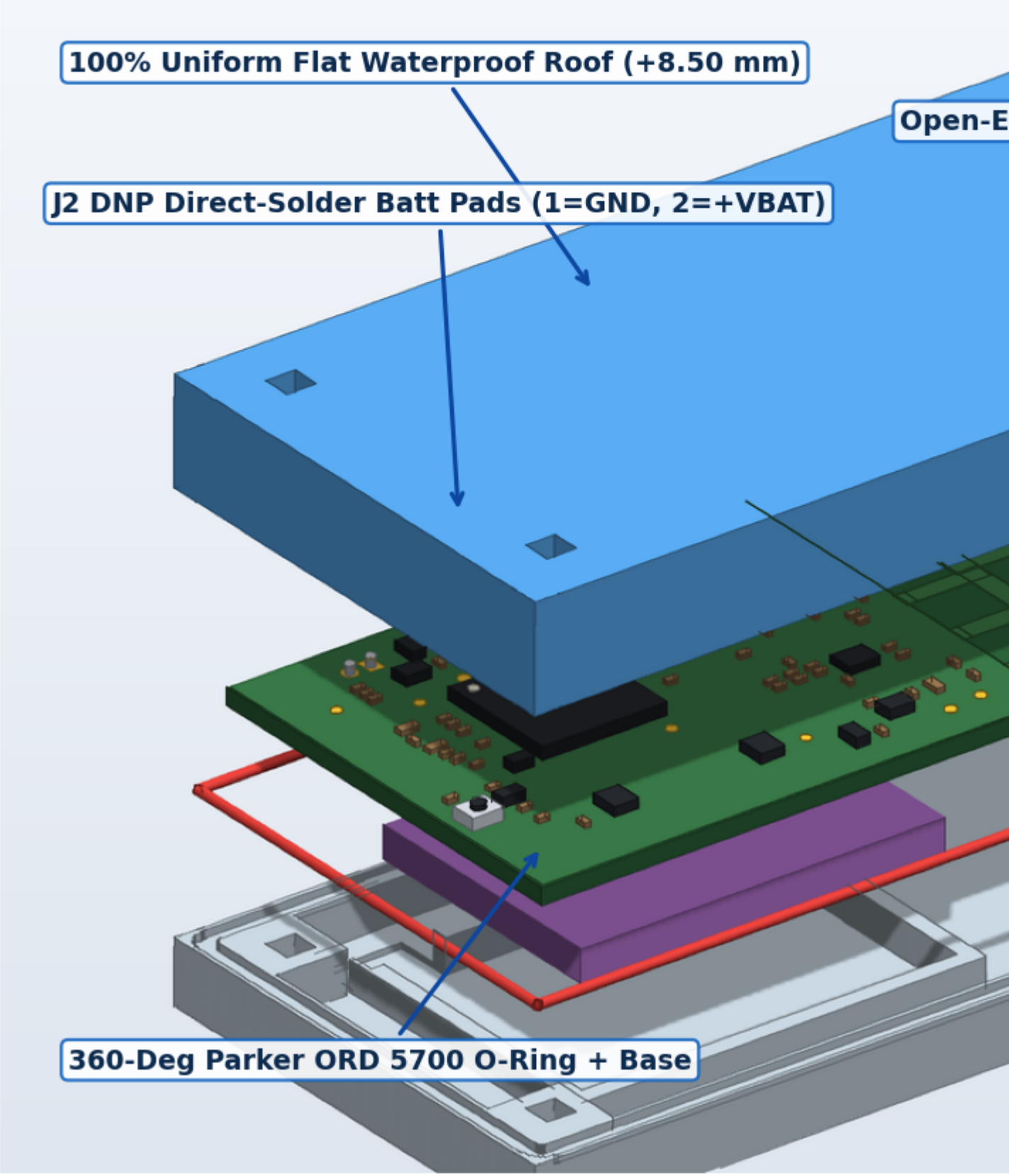

+
– D_VBAT
J2
Q3
U3
Q2
TP3
RAK
RAK3172LP-SiP
305049 EU868
STM32WLE5JC
BOOT
J3
U1
L_SMPS
D_RST
SW1
D_UA
SWD
D_SWD
J5
J4


D_VBAT
Q3
U3
J2
Q2
BOOT
J3
L_SMPS
RAK3172LP-S
305049
STM32WLE

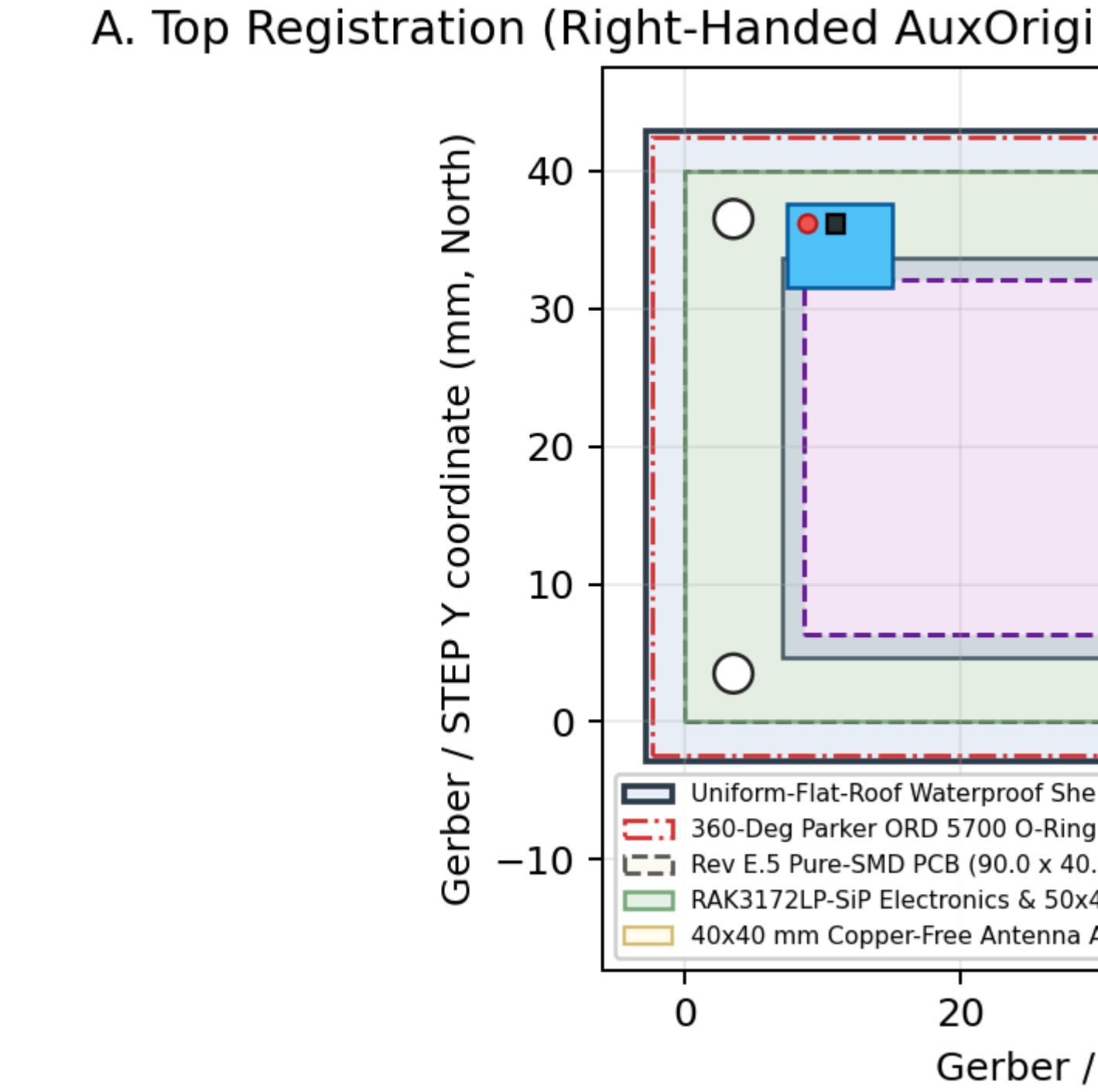


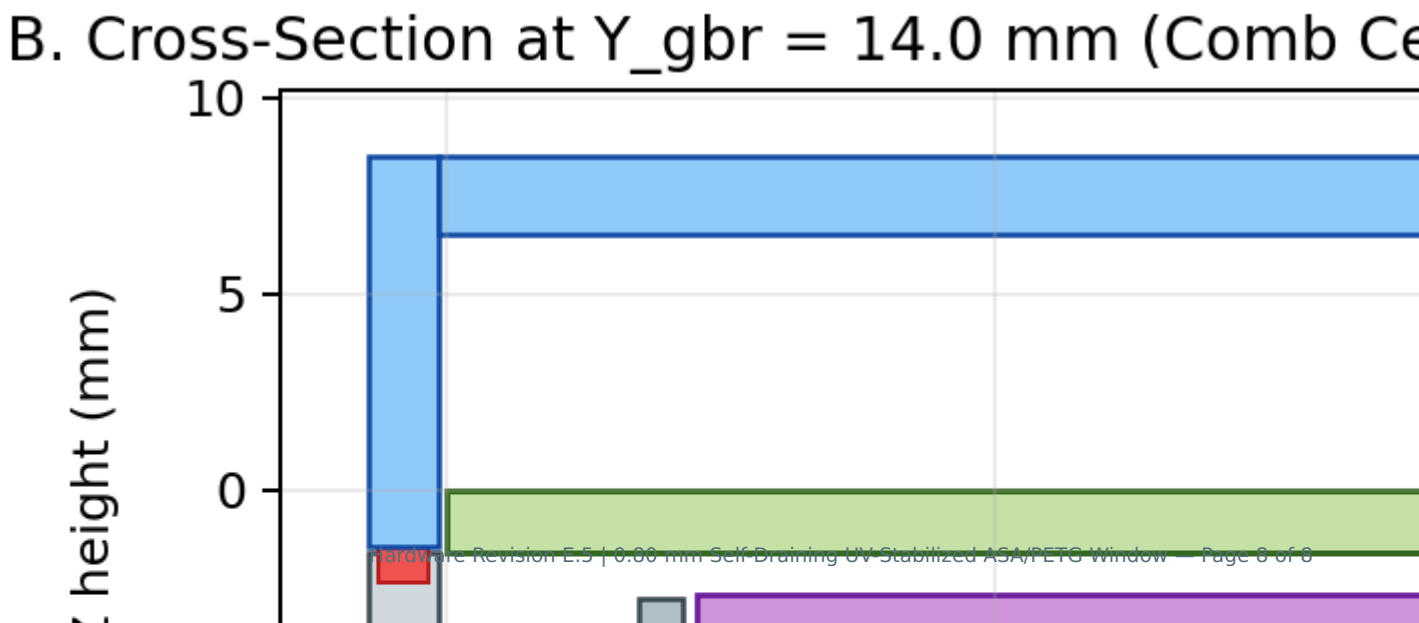
ISAC_868 Rev E.5 — Self-D

A. Exploded Assembly (Flat Lid + 3M 467MP + Pu



C. Longitudinal Cutaway @ Y_gbr = 14.0 mm (Open


A. Top Registration (Right-Handed AuxOrigi
Gerber / STEP Y coordinate (mm, North)
40
30
20
10
0
−10
0
20
Gerber /
Uniform-Flat-Roof Waterproof She
360-Deg Parker ORD 5700 O-Ring
Rev E.5 Pure-SMD PCB (90.0 x 40.
RAK3172LP-SiP Electronics & 50x4
40x40 mm Copper-Free Antenna A


B. Cross-Section at Y_gbr = 14.0 mm (Comb Ce
10
5
0
height (mm)